

Elevator controlled by combined logic

Equipment

TwinCAT 3 on a Windows computer

Preparation

- Download the project "TwinCAT Project Virtual Elevator" from Canvas
- Click on the file "TwinCAT Project Virtual Elevator.sln" to launch the project
- Start and select <UmRT_Default>
- Click on "Activate Configuration"  and then answer OK to download the configuration to your runtime PLC.

NOTE! In this project, the input and the output variables are declared in the Global Variable List (GVL). You must enter GVL. in front of each variable to access them. In many industrial projects, all inputs and outputs are declared in the global variable list, enabling access to the I/Os from all POU's.

Task 1

Write a control program in Functional Block Diagram (FBD).

Each level lamp should illuminate when the corresponding button is pressed. The elevator should go to the illuminated level, and the light should turn off. For that, a self-holding function is needed for each lamp. The lamps and buttons inside the elevator should function the same as those outside. You need three self-holding circuits, one for each level. The inside and outside level buttons are both connected to the OR of the self-holding circuits. Use the level sensors to turn off the self-holdings.

Task 2

Modify the code so that it stops when the red button is pressed. The elevator should start and continue to go to the floor it was heading to when the green button is pressed. For that, you need a new self-holding of a new mxRun variable and rename the two qx variables for the motor's up and down movements to two new mx instead. Use the new motor variables AND mxRun to assign the qx variables for the motor. This also makes it easier to implement the timers you need in task 3.

Task 3

The elevator should always stop briefly when it arrives at a new level.

Task 4

Open the door when the elevator arrives and close it before it starts.

Hand in the final solution to Canvas.